Ziang Xiao

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EMPLOYMENT

Johns Hopkins University
Assistant Professor, Computer Science Jan 2024

Microsoft Research, Montréal
Postdoctoral Researcher Sept 2022 - Jan 2024

Johns Hopkins University
Research Assistant Professor, Computer Science July 2022 - Jan 2024

EDUCATION

University of Illinois Urbana-Champaign 2023
Ph.D. in Computer Science
Committee: Hari Sundaram (Advisor), Karrie Karahalios (Advisor), Michelle X. Zhou, Heng Ji, and Brent W. Roberts

University of Illinois Urbana-Champaign 2016
B.S. with Highest Distinction in Psychology and High Distinction in Statistics & Computer Science
Dissertation Advisor: Dov Cohen

TEACHING

EN.601.490/690 Intro to Human Computer Interaction [Fall 2024-2026]

EN.601.790 Advance Human-Computer Interaction: Research Methods [Spring 2025-2026]

EN.601.792 Advanced Topics in Conversational User Interface [Spring 2024]

PUBLICATIONS

*: Equal authorship; †: Student mentee;

Conference Papers:

1. Jiang, H.†, Zhang, S., Zhu, D., Bai, Y., Truong, S. T., Yi, X., Koyejo, S., Xie, X., **Xiao, Z.** (2026) AI Evaluation Should Require Standardized Item-Level Data Releases. *Conference on Neural Information Processing Systems (NeurIPS), Position Paper.*
2. Sharma, N.†, Marchisio, K., Murray, K., **Xiao, Z.** (2026) The Narrative Gap: Can LLMs Help Us Navigate Diverse Narratives Across Languages? *Conference on Neural Information Processing Systems (NeurIPS), Evaluation and Dataset.*
3. Jiang, H.†, Kwon, S., Luo, J., **Xiao, Z.**, Zhang, S. (2026) Can We Trust Item Response Theory for AI Evaluation? *Conference on Neural Information Processing Systems (NeurIPS), Evaluation and Dataset.*

4. Li, Y.[†*], Zou, H.[†*], Wang, B., **Xiao, Z.** (2026) Contextualized Evaluation of Vision Language Models through Dynamic Interviews. *Conference on Neural Information Processing Systems (NeurIPS), Evaluation and Dataset.*
5. Hu, Z., Lian, J., Xiao, Z., Xiong, M., Wang, T., Xiao, T., Lei, Y., Jiang, F., Ding, K., **Xiao, Z.**, Yuan, N. J., Xie, X., Poovendran, R. (2026) Population-Aligned Persona Generation for LLM-based Social Simulation. *Conference on Neural Information Processing Systems (NeurIPS).*
6. Cui, C. Z., Yuan, X., **Xiao, Z.**, Ammanabrolu, P., Côté, M.-A. (2026) TALES: Text Adventure Learning Environment Suite. *Conference on Neural Information Processing Systems (NeurIPS), Evaluation and Dataset.*
7. Gao, J.[†], Sun, K., Huang, J.-t., Van Koevering, K., Ji, S., Huang, H., Shi, W., Lu, Z., **Xiao, Z.**, Khashabi, D., Dredze, M. (2026) How to Interpret Agent Behavior. *Conference on Neural Information Processing Systems (NeurIPS), Evaluation and Dataset.*
8. Ivey, J.[†], Field, A., **Xiao, Z.** (2026) What Makes a Good Response? An Empirical Analysis of Quality in Qualitative Interviews. *Conference on Empirical Methods in Natural Language Processing (EMNLP).*
9. Lu, Z., Yu, Y., Li, Z., Meng, Y., Jiang, N., Zang, C., Gao, J.[†], **Xiao, Z.** (2026) Using Grounded Theory for Agent Behavior Analysis at Scale. *Findings of the Association for Computational Linguistics: EMNLP.*
10. Li, Z., Ma, Z., He, S., Lu, Z., **Xiao, Z.**, Yin, M. (2026) Adaptive Selection of Effective AI Assistance in AI-assisted Decision Making Using Multi-Armed Bandits. *ACM Conference on Human Computation and Crowdsourcing (HCOMP).*
11. Pineda, K., Mishra, A., Chien, B., Guo, A., Williams, T., **Xiao, Z.**, Huang, C.-M. (2026) From Small Talk to Rapport: Exploring Robot Self-Disclosure in Collaborative Tasks. *IEEE International Conference on Robot and Human Interactive Communication (RO-MAN).*
12. Jiang, H.[†], Zhu, D., Yi, X., **Xiao, Z.**, Wei, Z.,, Xie, X.(2026) PICACO: Pluralistic In-Context Value Alignment of LLMs via Total Correlation Optimization. *International Conference on Machine Learning (ICML).*
13. Zhang, Z. S., Choi, Y., **Xiao, Z.**, Liao, Q. V., Zhao, J. (2026) Hi Alex or Dear Dr. Morgan?: Exploring How AI-Suggested Politeness Strategies Influence Email Writing and Social Perception Among Native and Non-Native Speakers. *ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW).*
14. Sharma, N.[†], Zhang, Z., Chen, H., Lee, D., Krishnan, N., Ren, G., **Xiao, Z.**, Li, Y. (2026) Feedback by Design: Understanding and Overcoming User Feedback Barriers in Conversational Agents. *To appear at the 2026 CHI Conference on Human Factors in Computing Systems (CHI)*
15. Van der Maden, W., Sadek, M., **Xiao, Z.**, Mottelson, Aske., Liao, Q. V., Zhu, J. (2026) Results-Actionability Gap: Understanding How Practitioners Evaluate LLM Products in the Wild. *To appear at the 2026 CHI Conference on Human Factors in Computing Systems (CHI)*
16. Tang, J., Chen, C., Li, J., Zhang, Z., Guo, B., Khalilov, I., Gebreegziabher, S., Yao, B., Wang, D., Ye, Y., Li, Y., **Xiao, Z.**, Yao, Y., Li, T. (2026) Dark Patterns Meet GUI Agents: LLM Agent Susceptibility to Manipulative Interfaces and the Role of Human Oversight. *To appear at the 2026 CHI Conference on Human Factors in Computing Systems (CHI)*
17. Wang, P., Zou, H.[†], Jiang, H.[†], Chen, H., Sun, T., Yi, X.,**Xiao, Z.**[*], Oswald, F.[*] (2026) Generative Personality Simulation via Theory-Informed Structured Interview. *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*
18. Zou, H.[†], Wang, P., Yan, Z., Sun, T., **Xiao, Z.** (2025) Can LLM Self-report?: Evaluating the Validity of Self-report Scales in Measuring Personality Design in LLM-based Chatbots. *Conference on Language Modeling (CoLM)*

19. Hyk, A., McCormick, K., Zhong, M., Ciucă, I., Sharma, S., Wu, J., Peek, J., Iyer, K., **Xiao, Z.**, Field, A. (2025) From Queries to Criteria: Understanding How Astronomers Evaluate LLMs. *Conference on Language Modeling (CoLM)*

20. Jiang, H.†, Yi, X., Wei, Z., **Xiao, Z.**, Wang, S., Xie, X.(2025) Raising the Bar: Investigating the Values of Large Language Models via Generative Evolving Testing. *International Conference on Machine Learning (ICML).*

21. Cao, S., Moon, J., Mahmood, A., Antony, V., **Xiao, Z.**, Liu, A., and Huang, C. (2025). Interruption Handling for Conversational Robots. *Robotics: Science and System (RSS).*

22. Sharma, N.†, Murray, K., and **Xiao, Z.** (2025). Faux Polyglot: A Study on Information Disparity in Multilingual Large Language Models. *In Proceedings of the 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL).*

23. Feng, F, Liao, Q. V., **Xiao, Z.**, Wortman Vaughan, J, Amy X. Zhang, A. X., and McDonald, D. W. (2025). Canvil: Designerly Adaptation for LLM-Powered User Experiences. *In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI)*

24. Li, Z., Zhu, H., Lu, Z., **Xiao, Z.**, and Yin, M.. (2025). From Text to Trust: Empowering AI-assisted Decision Making with Adaptive LLM-powered Analysis. *In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI)*

25. Pang, R. Y., Schroeder, H., Smith, K. S., Barocas, S., **Xiao, Z.**, Tseng, E., and Bragg, D.. (2025). Understanding the LLM-ification of CHI: Unpacking the Impact of LLMs at CHI through a Systematic Literature Review. *In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI)*

26. Pu, K., Lazaro, D., Arawjo, I., Xia, H., **Xiao, Z.**, Grossman, T., and Chen, Y.. (2025). Assistance or Disruption? Exploring and Evaluating the Design and Trade-offs of Proactive AI Programming Support. *In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI)*

27. Pitis, S., **Xiao, Z.**, Le Roux, N., and Sordoni, A. (2024). Improving Context-Aware Preference Modeling for Language Models. *Thirty-eight Conference on Neural Information Processing Systems (NeurIPS).*

28. Liu, Y.†, Blodgett, S. Cheung, J., Liao Q. V. , Olteanu A., and **Xiao, Z.**. (2024). ECBD: Evidence-Centered Benchmark Design for NLP. *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*

29. Wang, R., Todd, G., **Xiao, Z.**., Yuan, X, Côté, M. A., Clark, P., Jansen, P. (2024). Can Language Models Serve as Text-Based World Simulators?. *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*

30. Sharma, N.†, Liao, Q. V., and **Xiao, Z.**. (2024). Generative Echo Chamber? Effects of LLM-Powered Search Systems on Diverse Information Seeking. *In Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI)* `[Best Paper Award]`

31. Sordoni, A., Yuan, X., Côté, M. A., Pereira, M., Trischler, A., **Xiao, Z.**, Hosseini, A, Niedtner, F, and Le Roux, N. (2023). Joint Prompt Training of Stacked LLMs using Variational Inference. *In Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS).*

32. **Xiao, Z.** *, Zhang, S. *, Lai, V., and Liao, Q. V. (2023). Evaluating Evaluation Metrics: A Framework for Analyzing NLG Evaluation Metrics using Measurement Theory. *In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP) pp. 10967-10982.*

33. Wang, R., Todd, G., Yuan, E., **Xiao, Z.**, Côté, M. A., and Jansen, P. (2023). ByteSized32: A Corpus and Challenge Task for Generating Task-Specific World Models Expressed as Text Games. *In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP) pp. 13455–13471.*

34. **Xiao, Z.**, Li, T. W., Karahalios, K., and Sundaram, H. (2023). Inform the Uninformed: Improving Online Informed Consent Reading with an AI-Powered Chatbot. *In Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (pp. 1-17).*

35. **Xiao, Z.**, Liao, Q. V., Zhou, M., Grandison, T., and Li, Y. (2023). Powering an AI Chatbot with Expert Sourcing to Support Credible Health Information Access. *In Proceedings of the 28th International Conference on Intelligent User Interfaces (IUI) (pp. 2-18)*
36. Ge, Y. *, **Xiao, Z.** *, Diesner, J., Ji, H., Karahalios, K., and Sundaram, H. (2022). What should I ask: A knowledge-driven approach for follow-up question generation in conversational surveys. *In Proceedings of the 37th Pacific Asia Conference on Language, Information and Computation (PACLIC 37)*
37. Wauck, H., Woodard, B. S., **Xiao, Z.**, Li, T. W., and Bailey, B. P. (2020). A Data-Driven, Player-Centric Approach to Evaluating Spatial Skill Training Games. *In Proceedings of the Annual Symposium on Computer-Human Interaction in Play (CHI Play) (pp. 349-361).* `[Honorable Mention]`
38. **Xiao, Z.**, Zhou, M. X., Chen, W., Yang, H., and Chi, C. (2020). If I hear you correctly: Building and evaluating interview chatbots with active listening skills. *In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (pp. 1-14).*
39. **Xiao, Z.**, Ho, P. S., Wang, X., Karahalios, K., and Sundaram, H. (2019). Should we use an abstract comic form to persuade? Experiments with online charitable donation. *Proceedings of the ACM on Human-Computer Interaction, 3(CSCW), 1-28.*
40. **Xiao, Z.**, Zhou, M. X., and Fu, W. T. (2019). Who should be my teammates: Using a conversational agent to understand individuals and help teaming. *In Proceedings of the 24th International Conference on Intelligent User Interfaces (IUI) (pp. 437-447).*
41. Gao, M. *, **Xiao, Z.** *, Karahalios, K., and Fu, W. T. (2018). To Label or not to Label: The Effect of Stance and Credibility Labels on Readers' Selection and Perception of News Articles. *Proceedings of the ACM on Human-Computer Interaction, 2(CSCW), 1-16.*
42. Chiu, P. T., Wauck, H., **Xiao, Z.**, Yao, Y., and Fu, W. T. (2018). Supporting spatial skill learning with gesture-based embodied design. *In 23rd International Conference on Intelligent User Interfaces (IUI) (pp. 67-71).*
43. **Xiao, Z.**, Wauck, H., Peng, Z., Ren, H., Zhang, L., Zuo, S., Yao, Y., and Fu, W. T. (2018, March). Cubicle: An adaptive educational gaming platform for training spatial visualization skills. *In 23rd International Conference on Intelligent User Interfaces (IUI) (pp. 91-101).*
44. Wauck, H., **Xiao, Z.**, Chiu, P. T., and Fu, W. T. (2017). Untangling the relationship between spatial skills, game features, and gender in a video game. *In Proceedings of the 22nd International Conference on Intelligent User Interfaces (IUI) (pp. 125-136).*
45. **Xiao, Z.**, Yao, Y., Yen, C. H., Dey, S., Wauck, H., Leake, J. M., Woodard, B., Wolters, A., and Fu, W. T. (2017). A Scalable Online Platform for Evaluating and Training Visuospatial Skills of Engineering Students. *In 2017 ASEE Annual Conference & Exposition.*
46. **Xiao, Z.** (2016). Does Grammatically Gendered Language Have an Impact on Investors' behaviors? Companies with Grammatically Masculine Names Show More Stock Momentum Than Those with Grammatically Feminine Names. *Bachelors dissertation, University of Illinois Urbana-Champaign.*

Journal Papers:

1. Chen, C., Zhang, Z., Guo, B., Ma, S., Khalilov, I., Gebreegziabher, S. A., Ye, Y., **Xiao, Z.**, Yao, Y., Li, T., Li, T. J.-J. (2026) The Obvious Invisible Threat: Vulnerabilities of LLM-Powered GUI Agents in Web-Interaction Tasks. *ACM Transactions on AI Security and Privacy (TAISAP).*
2. Zhao, Y., Qiu, X., **Xiao, Z.**, Kaufman, M.(2025). Trauma narratives and healing of offspring caregivers of breast cancer patients: a mixed qualitative methods study. *BMC Psychology*
3. **Xiao, Z.**, Mennicken, S., Huber, B., Shonkoff, A., and Thom, J. (2021). Let Me Ask You This: How Can a Voice Assistant Elicit Explicit User Feedback?. *Proceedings of the ACM on Human-Computer Interaction, 5(CSCW2), 1-24.*
4. Vaccaro, K., **Xiao, Z.**, Hamilton, K., and Karahalios, K. (2021). Contestability for content moderation. *Proceedings of the ACM on human-computer interaction, 5(CSCW2), 1-28.*

5. Goldstein, M. H., Froiland, J. M., **Xiao, Z.**, Woodard, B. S., Li, T., and Philpott, M. L. (2021). Application of online visual-spatial training to increase visual-spatial ability and growth mindset of engineering students. *Int. J. Eng. Educ, 37, 1674-1682.*
6. **Xiao, Z.**, Zhou, M. X., Liao, Q. V., Mark, G., Chi, C., Chen, W., and Yang, H. (2020). Tell me about yourself: Using an AI-powered chatbot to conduct conversational surveys with open-ended questions. *ACM Transactions on Computer-Human Interaction (TOCHI), 27(3), 1-37.*

Book Chapters:

1. Torkamaan, H., Tahaei, M., Buijsman, S., **Xiao, Z.**, Wilkinson, D., & Knijnenburg, B. P. (2024). The Role of Human-Centered AI in User Modeling, Adaptation, and Personalization—Models, Frameworks, and Paradigms. *In A Human-Centered Perspective of Intelligent Personalized Environments and Systems (pp. 43-83). Cham: Springer Nature Switzerland*

Workshop, Posters and Demos:

1. Huiqi Zou† Yijiang Li, and **Xiao, Z.** (2024). Towards Dynamic and Realistic Evaluation of Multi-modal Large Language Model *GenBench workshop: The second workshop on generalisation (benchmarking) in NLP at 2024 Empirical Methods in Natural Language Processing (GenBench@EMNLP2024)*
2. Liao, Q. V., and **Xiao, Z.** (2023). Rethinking Model Evaluation as Narrowing the Socio-Technical Gap. *AI & HCI Workshop at the 40th International Conference on Machine Learning (ICML)* `[Honorable Mention]`
3. Pitis, S., **Xiao, Z.**, and Sordoni, A. (2023). Canonical Design for Language Agents using Natural Language Reward Models. *1st Workshop on AI meets Moral Philosophy and Moral Psychology at Thirty-seventh Conference on Neural Information Processing Systems (MP2@NeurIPS).*
4. **Xiao, Z.**, Yuan, X., Liao, Q. V., Abdelghani, R., and Oudeyer, P. Y. (2023). Supporting Qualitative Analysis with Large Language Models: Combining Codebook with GPT-3 for Deductive Coding. *In Companion Proceedings of the 28th International Conference on Intelligent User Interfaces (IUI) (pp. 75-78).*
5. Li, T. W., **Xiao, Z.**, Goldstein, M. H., Philpott, M. L., and Woodard, B. (2021). Evaluating an Intelligent Sketching Feedback Tool for Scalable Spatial Visualization Skill Training. *In 2021 ASEE Virtual Annual Conference Content Access.*
6. Woodard, B. S., Li, T. W., **Xiao, Z.**, Goldstein, M. H., and Philpott, M. L. (2021). Work in Progress: Spatial Visualization Assessment and Training in the Grainger College of Engineering at the University of Illinois. *In 2021 ASEE Illinois-Indiana Regional Conference.*
7. Zhou, M. X., Chen, W., **Xiao, Z.**, Yang, H., Chi, T., and Williams, R. (2019). Getting virtually personal: chatbots who actively listen to you and infer your personality. *In Proceedings of the 24th International Conference on Intelligent User Interfaces: Companion (IUI) (pp. 123-124).*
8. Xiao, Z., Zuo, S., Zhao, J., Fu, W. T., Goldstein, M. H., Philpott, M. L., M., Laystorm-Woodard, J., Pool, M., Wolters, A., and Woodard, B. S. (2019). Understanding interrelated growth mind-set and academic participation & performance. *In 2019 ASEE Annual Conference & Exposition.*
9. **Xiao, Z.**, Yao, Y., and Fu, W. T. (2018). An Intelligent Educational Platform for Training Spatial Visualization Skills. *In Proceedings of the 23rd International Conference on Intelligent User Interfaces Companion (IUI) (pp. 1-2).*
10. **Xiao, Z.**, and Bub, K. 2016. Longitudinal Associations between Self-Regulation and Mental Health across Early Childhood to Adolescence. *16th Biennial Meeting of Society for Research on Adolescence (SRA 16)*

GRANTS

2026 Welch Center Transdisciplinary Research and Collaboration Grant
Utilizing AI to Promote SGLT-2 Inhibitor and GLP-1 Therapy Adherence and Safety (MPI; 2026 - 2028)

2026 DSAI Trusted Data for Trustworthy AI
OpenEval: Data Infrastructure for the Science of AI Evaluation (PI; 2026)

2025 Welch Center Transdisciplinary Research and Collaboration Grant
Promoting Hypertension Medication Adherence through AI (MPI; 2025 - 2027)

Google - Gemma Academic Program GCP Credit Award
Understanding multilingual information disparity in Generative AI (PI; 2025 - 2026)

2025 Microsoft Accelerating Foundation Models Research
Understanding and Improving AI Value Alignment Evaluation via a Measurement Theory Framework (PI; 2025)

2025 Johns Hopkins Discovery Award
Using AI to Make Social Media More Prosocial (Co-PI; 2025 - 2026)

Schmidt Sciences AI Safety Science
A Toolkit to Create Valid and Efficient Benchmarks with Measurement Theory (PI; 2025 - 2027)

Adobe Research Gift (PI; 2024)

Amazon Research Awards
Interactive and Contextualized Evaluation for Large Language Models (PI; 2024 - 2025)

2024 Johns Hopkins Discovery Award
Virtual Neurology Clinic: A Human-AI Interface for Remote Autonomous Augmented/Virtual Reality Neurologic Screening and Triaging (Co-PI; 2024 - 2025)

2024 SNF Agora Faculty Grant
AI-Curated Democratic Discourse (Co-PI; 2024 - 2025)

2024 SNF Agora Faculty Grant
Understanding and Mitigating Generative Echo Chamber in Conversational Search (PI; 2024 - 2025)

Google - Gemma Academic Program GCP Credit Award
Designing Conversational Search that Facilitates Diverse Information Seeking (PI; 2024 - 2025)

SERVICE

Organizing Committee:

- IUI 2023 [Publicity Chair]
- IUI 2025 [Workshop Chair]
- Humen-centric Generative AI @TiiS [Guest Editor]
- Human-Centered Evaluation and Auditing of Language Models @CHI 2024 [Organizer]
- Human-Centered Evaluation and Auditing of Language Models @CHI 2025 [Organizer]

- Human-Centered Evaluation and Auditing of Language Models @CHI 2026 [Organizer]
- 3rd NLP+HCI Workshop @NAACL 2024 [Co-Organizer]
- 4th NLP+HCI Workshop @EMNLP 2025 [Co-Organizer]
- Human-Centered Evaluation of Language Technologies @ EMNLP 2024 [Co-Insturctor]

Associate Chair:

- CHI 2024-27
- CHI 2021 Late Breaking Works
- CSCW 2021 Poster
- CSCW 2022
- IUI 2023, 2025
- WWW 2023
- FAccT 2023-25

Frequent Reviewer: CHI, CSCW, IUI, UIST, DIS, *ACL, CHI Play, ECSCW, ACM TOCHI, ACM TIIS, IJHCI, Nature Human Behaviour, Science

AWARDS & HONORS

- Joel Dean Excellence in Teaching Award
- 2025 Star Track Scholar - Microsoft Research Asia
- Outstanding Reviews [CHI 22-26, UIST 22, NeurIPS 26]
- Microsoft Global Hackathon Award Winner - Responsible AI (2023)
- Teacher Ranked as Excellent (2022)
- Graduate College Conference Participation Awards (2021)
- Harry Triandis Award (2016)
- James E. Spoor Scholarship (2016)